Chat Export for sanidhya.rana_contractor@celanese.com

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: cable glands

Assistant: To help identify the best material for your cable gland application, could you please clarify:

1. What is the exact use of the polymer in your cable gland? For example, is it for the main body, the sealing grommet, or another component?
2. Where will the cable gland be used? Is it for an industrial panel, outdoor junction box, consumer electronics, or another end-use scenario?

Your answers will help me understand the specific requirements and operating conditions for your application.

User: It is the main body

Assistant: Thank you for clarifying that the polymer is for the main body of the cable gland.

Could you please let me know:
1. Where will the cable gland be used? For example, is it for an industrial panel, outdoor junction box, electric vehicle, railway application, or consumer electronics?
2. If you're unsure, examples include: cable glands for industrial machinery, outdoor electrical installations, EV battery systems, or indoor switchboards.

This information will help me understand the operating environment and specific requirements for your application.

User: it will be in an industrial application and outdoor

Assistant: Thank you for specifying the industrial and outdoor use. To ensure the material meets your requirements, could you please provide:

1. What is the minimum wall thickness of the cable gland main body?
2. Do you have any preferred base materials for this application (such as polyamide, polycarbonate, or another polymer)?
3. Are you open to alternative material suggestions if your preferred material is not available or optimal?

These details will help identify the best material options for your cable gland.

User: 0.8 mm and open to suggestions

Assistant: Here are the key material requirements (CTQs) assumed for your cable gland main body in industrial outdoor use:

1. Minimum wall thickness: 0.8 mm
2. Flame rating: UL94 V-0 at 0.8 mm
3. Glow Wire Flammability Index (GWFI): 750°C at 0.8 mm
4. Flame retardant: Halogen and red phosphorus-free
5. UV/weather resistance: Yes
6. Processing requirement: Injection molding

Are all of these requirements correct for your application? Please confirm if these match your needs, or let me know if any should be changed.

User: looks good can you recommend the grades

Assistant: Based on your requirements for an industrial outdoor cable gland main body, the following grades may be suitable options:

1. FRIANYL® A3 H GF25 V0 NC 1102/EA (PA66, 25% glass fiber)
2. FRIANYL® A3 GF25 V0 NC 1102 (PA66, 25% glass fiber)
3. FRIANYL® A3 GF30 V0 (PA66, 30% glass fiber)
4. CELANEX® XFR 6842 GF30 (PBT, 30% glass fiber)
5. CELANEX® XFR 6852 GF30 (PBT, 30% glass fiber)
6. FRIANYL® A3 GF35 V0 (PA66, 35% glass fiber)

All these grades offer UL94 V-0 at 0.8 mm, GWFI ≥750°C, are halogen/red phosphorus-free, UV/weather resistant, and suitable for injection molding. The PA66 grades are widely used for cable glands, while the CELANEX® PBT grades provide enhanced chemical and hydrolysis resistance for harsh environments.

Please note: These are potential options based on technical requirements. Final material selection and product design decisions should be validated for your specific application and compliance needs. If you have specific color or chemical resistance requirements, let me know for further refinement.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.